

6 Engineering Desktop Client

ACTIVE | TOTAL: orig ~4d · forecast ~1d | PROGRESS: spent ~0d · rem ~1d | forecast end 12 Oct 2026

GOAL

Evolve the existing JavaFX Development Client into the project's full Engineering Desktop Client while preserving its public-interface-only boundary to SI-01.

RESULT

- SI-02 is defined as the maintainable Engineering Desktop Client software item.
- Normal operator interaction remains IF-04 Web.
- SI-02 remains independent from SI-01 implementation classes/files and keeps
- The Java 21 / BentoFX workbench and Windows app-image path are implemented and

END DEMO

- Start the Engineering Client and connect it to a running SI-01.
- Show connection/synchronisation state, current TimingNode status, registrations,
- Exercise one supported engineering command/simulation and inspect the result.
- Disconnect/reconnect and rebuild the authoritative view.
- Using SI-02, revoke through the trash action,
- Show Device Log / Client Log / Remote Shell independently where available.

DOCUMENTATION

- 30-UC R-85
- 41-02-SSD W-45
- 50-SDE-03 W-80
- 32-03-IDD R-85
- 60-SVP R-75

DOCUMENTATION / DECISIONS

D01

DONE

Establish first SI-02 Engineering Client requirement slice

Establish SI02-REQ-001..008...

APPLICATION / PRODUCT

D02 after D01

DONE

Select Engineering Client desktop stack, client boundary and packaging

Select Java 21, JavaFX 21,...

A01 after D01 D02

DONE

Restructure the existing Engineering Client around the selected workbench/services

Completed through Java issue...

VERIFICATION / TEST

V01 after A01

NEXT

Verify Engineering Client connection, sync/reconnect and representative workbench behaviour

Verify connection/status...

PLANNING CHANGES

- 2026-10-09 — Complete A01 through Java PR #402: SI-02 now runs on Java 21 / JavaFX 21 with BentoFX, an instance-scoped system context, real dockable panes and a CI-verified Windows jpackage app-image. V01 is next; with only that running-system verification remaining, the Step-6 remainder is re-estimated at 1 project day.
- 2026-10-09 — Classify the reusable Engineering Desktop Client as SI-02. Keep UC-009 as its system use case, IF-03 as its primary SI-01 boundary and IF-04 Web as the normal operator path; UC-008 remains outside the current baseline.
- 2026-10-09 — Close Step 5 on SI-01/server evidence and assign the real GUI revoke/DELETED/reconnect/recovery scenario to Step 6 V01, releasing A01.
- 2026-10-09 — Select Java 21, JavaFX 21, BentoFX 0.16.0, JDK HTTP/WebSocket, Jackson, Maven and a jpackage app-image as the Step-6 desktop baseline.

PLANNING CHANGES (CONT.)

- 2026-10-09 — Reframe Step 6 from an operator desktop GUI to an Engineering Desktop Client; normal operator use remains IF-04 Web.
- 2026-10-07 — Promote the first SI-02 Draft requirement slice before selecting GUI technology; implementation remains gated on the then-current Step-5 V06 plan.